

Goktug Aygun

+32494199424 | goktug.aygun@student.kuleuven.be | linkedin.com/in/goktugaygun | github.com/goktug-aygun | goktugaygun.com
Andreas Vesaliusstraat, Leuven / Belgium

EDUCATION

KU Leuven, Leuven, Belgium

MEng in *Computer Science, Focusing on Artificial Intelligence*

Sep 2025 -Exp. Jun 2027

Coursework: *Natural Language Processing, Genetic Algorithms and Evolutionary Algorithms, Constraint Solving, Distributed Systems, Declarative Languages, Computer Vision, Reinforcement Learning, Information Retrieval and Search Engines, Modelling of Complex Systems, Data Mining*

Sabanci University, Istanbul, Turkey

BSc in *Computer Science and Engineering*, Minor in *Decision & Behavior*

Sep 2021 - Jun 2025

GPA: 3.78 / 4.00

Ranking: 26 / 319

Coursework: *Data Structures & Algorithms, Theory of Computation, Discrete Math, Computer Architecture, Artificial Intelligence, Machine Learning, Operating Systems, Database Systems, Computer Networks, Software Engineering*

Sakip Sabanci Encouragement Award, 2022 & 2023: Full Tuition Scholarship Awarded for Academic Excellence

TECHNICAL EXPERIENCE

DidactLabs, *Software Developer Intern*, Leuven, Belgium

Feb - May 2026

- Contributed to the backend development of an AI-integrated education platform using TypeScript
- Assisted with feature implementation, bug fixes, and codebase refactoring
- Collaborated with the engineering team to maintain code quality and support ongoing development efforts

OBSS, *Software Developer Intern*, Istanbul, Turkey

Jun – Sep 2024

- Developed full-stack application using React.js and .NET
- Designed REST APIs with authentication and authorization mechanisms

Sabanci University, *Undergraduate Learning Assistant*, Istanbul, Turkey

Sep 2022 – Jan 2024

- Mentored 2000+ students in Python and C++ across multiple semesters
- Led weekly recitation sessions and office hours to reinforce course concepts

PROJECTS

Cloud-Deployed Service-Oriented Distributed Ticketing System

Sep 2025 – Jan 2026

- Designed and deployed a fault-tolerant, service-oriented ticketing platform on Google Cloud
- Implemented retry and timeout mechanisms to integrate reliable and unreliable providers

Optimization of the Travelling Salesman Problem Using Genetic Algorithms

Sep 2025 – Jan 2026

- Implemented a Genetic Algorithm solver for the Travelling Salesman Problem using established evolutionary operators
- Investigated trade-offs between convergence speed and solution optimality across different configurations
- Visualized evolutionary progress to analyze performance trends over generations

Analyzing Sabanci University Curricula in Relation to the Sustainable Development Goals

Nov 2024 – Jun 2025

- Leveraging keyword based classification and LLMs to investigate the alignment between SDGs and course curricula

Designing and Implementing an LLM-based Course Assistant, Graduation Project (ENS491) Sep 2024 – Jun 2025

- Creating an LLM-based assistant, aiming to assist students and assistants in Sabanci University
- Adapting LLaMA, creating available and trustworthy AI based assistant using Retrieval Augmented Generation

Flight Roster Project, Software Engineering Course (CS308) Feb – Jun 2024

- Built a web application simulating airline operations, architecting the backend and developing RESTful APIs using Node.js while leading a backend team of 7
- Designed and deployed MySQL databases on AWS Cloud Services and leveraged MongoDB for NoSQL solutions, ensuring reliable and efficient data storage and retrieval

Bug Severity Detector Project, Machine Learning Course (CS412) Feb – Jun 2024

- Created a classifier utilizing different models with a team of 4
- Aimed to detect the severity of a given bug in range [1,7] based on given textual description

Investigating Graph Coloring Project, Algorithms Course (CS301) Feb – Jun 2024

- Collaborated with a team of 2 to investigate graph coloring problem with various heuristic and greedy algorithms
- Implemented and benchmarked heuristic and greedy algorithms for graph coloring

Study Buddy Mobile Application Project, Mobile Application Development Course (CS310) Sep 2023 – Jan 2024

- Built a collaborative learning application connecting students enrolled in similar courses, enabling posting and course filtering features
- Designed and developed the Android front-end and Java/Spring-based backend, containerizing services with Docker

DiSUCord - Message Networking via Channels, Computer Networks Course (CS408) Sep 2023 – Jan 2024

- Employed TCP sockets and designed server application layer for users to interact with using C# for Graphic User Interfaces besides .NET for necessary infrastructure
- Developed an interactive communication platform for students

Personal Portfolio Website Jun 2023 – Jul 2024

- Created a personal website, goktugaygun.com, using HTML, CSS, JS, React and Bootstrap framework

Analysis of Female MPs in Turkish Assembly, Introduction to Data Science Course (CS210) Jan – Jun 2023

- Scraped and analyzed data from the Grand Assembly of Turkey to investigate gender inequality, leading a team of 5
- Generated visualizations and reports using Matplotlib and Seaborn to communicate key analytical findings

Deaths Around the World by Cause Database Application, Database Systems Course (CS306) Jan – Jun 2023

- Designed and built relational models and database system to store real world data obtained from ourworldindata.org and investigated it using Python with visualization tools

PROFESSIONAL SKILLS

- **Programming Languages:** Python, C++, C#, Java, JavaScript, SQL
- **Web & Backend Technologies:** REST APIs, Flask, Spring, .NET, Node.js, React.js
- **Databases:** MySQL, MongoDB, NoSQL, Chroma, Firebase
- **Cloud & DevOps:** Google Cloud, Docker, Git
- **AI & ML:** Langchain, LLMs, RAG, NLP, Pandas, Matplotlib, Selenium, Hugging Face, PyTorch, Scikit-Learn
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication, Team Collaboration, Time Management
- **Languages:** Turkish (Native), English (Advanced, Scored 113 in TOEFL), German (Beginner)

REFERENCES

Available upon further request